

Predicting Chess Puzzle Difficulty with Transformers

1st Szymon Miłosz

*Institute of Applied Computer Science
Lodz University of Technology
Lodz, Poland
szymonmilosz99@gmail.com*

2nd Paweł Kapusta

*Institute of Applied Computer Science
Lodz University of Technology
Lodz, Poland
pawel.kapusta@p.lodz.pl*

Abstract—This study addresses the challenge of quantifying chess puzzle difficulty - a complex task that combines elements of game theory and human cognition and underscores its critical role in effective chess training. We present GlickFormer, a novel transformer-based architecture that predicts chess puzzle difficulty by approximating the Glicko-2 rating system. Unlike conventional chess engines that optimize for game outcomes, GlickFormer models human perception of tactical patterns and problem-solving complexity.

The proposed model utilizes a modified ChessFormer backbone for spatial feature extraction and incorporates temporal information via factorized transformer techniques. This approach enables the capture of both spatial chess piece arrangements and move sequences, effectively modeling spatio-temporal relationships relevant to difficulty assessment.

Experimental evaluation was conducted on a dataset of over 4 million chess puzzles. Results demonstrate GlickFormer’s superior performance compared to the state-of-the-art ChessFormer baseline across multiple metrics. The algorithm’s performance has also been recognized through its competitive results in the IEEE BigData 2024 Cup: Predicting Chess Puzzle Difficulty competition, where it placed 11th.

The insights gained from this study have implications for personalized chess training and broader applications in educational technology and cognitive modeling.

Index Terms—Chess puzzles, Transformer, Deep learning, Temporal modeling

I. INTRODUCTION

Chess problems, or puzzles, have long been an essential part of chess training, designed to help players develop their tactical abilities by finding the best sequence of moves in a given position. These puzzles, derived from historic games or carefully crafted positions, can range from simple one-move solutions to complex multi-move combinations. However, assessing the difficulty of chess problems is not a straightforward task, as the factors influencing difficulty extend beyond basic game-theory principles. While the rules of chess and game objectives are well-defined, the complexity of a puzzle depends largely on human perception—players must identify patterns, recognize tactical motifs, and mentally navigate through a multitude of potential moves. This complexity underscores the importance of accurately evaluating puzzle difficulty levels, a task that is far from artificial. Indeed, it is crucial for chess players to engage with puzzles that match their skill level, ensuring they can effectively improve their abilities

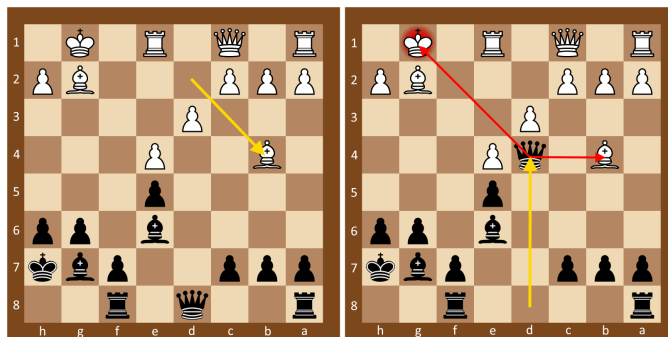
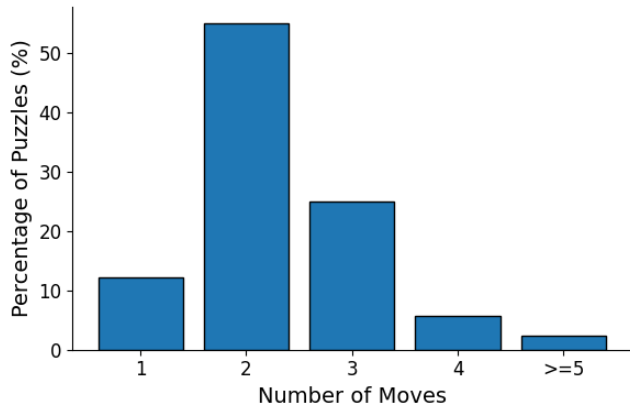


Fig. 1: Example of a simple two-move chess puzzle; White played **Bb4** attacking Black’s rook which is a blunder (left). The goal of the puzzle is to find the **Qd4+!** move, a fork attacking both the White king and bishop (right). In the next move White is forced to move the king or block the check, leaving the bishop undefended. Black then captures the unprotected bishop with **Qxb4**, resolving the puzzle by gaining material through the capture.

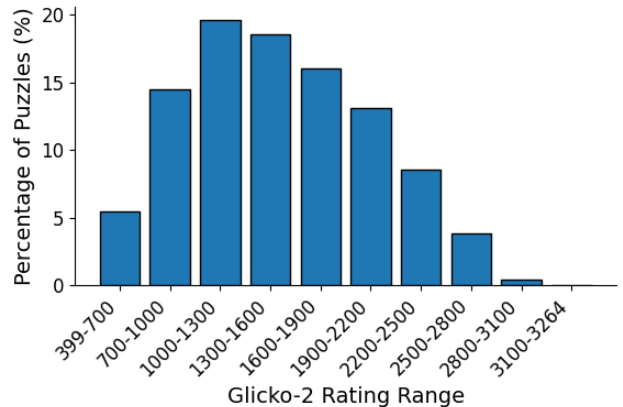
through appropriately challenging tasks. The need for such tailored training highlights why assessing puzzle difficulty is a significant and necessary aspect of chess education and improvement.

Chess rating systems, such as Elo [1], Glicko [2], and Glicko-2 [3], have long been used to quantify player strength. These systems provide a mathematical framework for estimating relative skill levels based on game outcomes. While these rating systems are well-established for player evaluation they offer insights into the challenge of assessing puzzle difficulty. However, this approach reveals a significant practical limitation: to obtain a reliable rating, each puzzle would need to be solved hundreds of times by players of varying skill levels. This resource-intensive process, while theoretically sound, proves impractical for large-scale puzzle evaluation or for assessing newly created challenges.

The task of predicting chess puzzle difficulty presents a unique challenge compared to traditional chess move prediction. In standard chess engines, the goal is to identify the move that maximizes the probability of winning, given a well-defined set of rules and a clear win condition. In contrast,



(a) Distribution of the required number of moves to solve the puzzle in the dataset



(b) Distribution of puzzle difficulty in the dataset

Fig. 2: Distributions of selected dataset statistics

assessing puzzle difficulty requires modeling the abstract concept of how players with varying skill levels perceive and solve tactical patterns.

Transformer architectures, which have revolutionized natural language processing and other domains, offer a promising approach to this challenge. Their ability to capture long-range dependencies and process sequential data makes them well-suited for analyzing chess positions and move sequences. Notably, the success of chess-specific transformers like ChessFormer [4], utilized in the Leela Chess Zero chess engine, demonstrates the potential of these architectures in chess-related tasks.

In this paper, we present GlickFormer, a transformer-based model designed specifically for predicting chess puzzle difficulty by approximating the rating obtained with Glicko-2 system. By leveraging the power of attention mechanisms and the ability to capture complex patterns across both the spatial arrangement of chess pieces and the temporal sequence of moves, GlickFormer aims to provide a more accurate, consistent, and scalable method for assessing puzzle difficulty.

II. RELATED WORK

Chess problem difficulty estimation is a relatively underexplored area in both artificial intelligence and chess research. Although considerable effort has been invested in developing chess engines capable of outperforming human grandmasters, much less attention has been given to understanding the cognitive challenges involved in solving chess puzzles. It is a non-trivial task to even define the difficulty of the problem. Previous work has shown, that assessing chess puzzle difficulty is hard even for human experts [5]. However, by following the suggestions of Grandmasters, some progress have been made on automatic problem difficulty assessment by mimicking the thought process of human players and constructing the meaningful search trees [6] [7].

Chess engines like Stockfish and Leela Chess Zero (LCZero) have already demonstrated the power of game-tree

search and neural networks to master chess at a superhuman level. LCZero, in particular, has shown evidence of a learned "look-ahead" strategy, as highlighted in recent research [8]. These models are trained to maximize winning probabilities and evaluate positions based on outcomes derived from large-scale self-play. However, GlickFormer diverges from these objectives. Instead of seeking optimal moves, it focuses on understanding the cognitive complexity behind problem-solving. In doing so, GlickFormer is designed to predict how difficult a puzzle will be based on how human players process information rather than how a machine would determine the best move.

The advent of transformers, as outlined in the "Attention is All You Need" paper [9], has revolutionized the field of artificial intelligence. Transformers excel at processing sequential data by employing a self-attention mechanism, which allows them to capture long-range dependencies and model intricate relationships in the data. This makes them particularly suited for tasks that involve both spatial and temporal components, such as video analysis or, in this case, chess puzzles. Recent advancements, such as the Video Vision Transformer (ViViT) [10], have demonstrated the ability of transformers to model complex patterns over time in non-Euclidean spaces [11], making them an ideal choice for predicting the difficulty of chess puzzles. Furthermore, a recent study [12] has shown that large-scale transformer models trained on extensive chess datasets can achieve remarkable performance in chess, reaching a Lichess blitz Elo of 2895 against humans and solving challenging chess puzzles without explicit search algorithms. These developments underscore the potential of transformers in modeling complex game scenarios and predicting the difficulty of chess puzzles.

In a recent development, a new study has emerged [13], which participated in the same IEEE BigData 2024 Cup competition [14] as ours. It employs a combination of Convolutional Neural Networks (CNN) and bidirectional Long

Short-Term Memory (LSTM) networks. While our approach leverages transformer architecture, this study opts for more traditional deep learning methods, while still incorporating temporal information. The CNN-LSTM approach achieved comparable, albeit lower, results on the public leaderboard in both preliminary and final scores, compared to our transformer-based solution. This parallel development in the field underscores the complexity of chess rating estimation and the diversity of potential approaches. While the CNN-LSTM model demonstrates the viability of traditional deep learning methods for this task, the superior performance of our transformer-based architecture suggests that the self-attention mechanism may be particularly well-suited for capturing the nuanced patterns in chess problem difficulty.

III. DATASET

The dataset consists of 4.2 million chess puzzles from Lichess.org, covering a wide range of difficulties from beginner to grandmaster levels (Figure 2). Each puzzle includes:

- The starting position in Forsyth-Edwards Notation (FEN).
- The sequence of moves leading to the solution.
- The Glicko-2 rating r , indicating puzzle difficulty.
- The rating deviation RD , reflecting assumed reliability in the rating.
- Additional metadata such as themes, number of plays, popularity scores, and associated game tags.

The mean and standard deviation of Glicko-2 rating values in the dataset are 1516 and 543 respectively. Majority of the Rating Deviation values in the dataset fall between 80 and 90, indicating a relatively high confidence in the difficulty ratings of these puzzles. However, puzzles that have been solved less frequently exhibit larger rating deviations, often exceeding 200. This higher RD reflects greater uncertainty in the estimated difficulty due to limited player interactions with the puzzles.

IV. METHODOLOGY

In this section, we describe our proposed approach for predicting chess puzzle difficulty. We begin with the puzzle encoding scheme that captures the spatial and temporal aspects of chess puzzles. We then detail our model architecture, which includes spatial feature extraction using the ChessFormer backbone and the incorporation of temporal information through two variants of our proposed model.

A. Puzzle Encoding

Chess puzzles are encoded as sequences of N boards, where each board is represented as $B_n \in \mathbb{R}^{16 \times 8 \times 8}$. This encoding draws inspiration from AlphaZero’s [15] approach, where similar board representation is used to capture the game state. The entire puzzle encoding is given by:

$$\mathbf{P} = \{B_1, B_2, \dots, B_N\}.$$

This encoding captures the intricate relationships between moves while maintaining a comprehensive representation of the chessboard and its pieces.

Each B_n is a binary tensor with 16 channels over an 8×8 grid, organized as follows:

- **Piece Representation (12 channels):**

- **6 Channels for Mover’s Pieces:** Encoding the presence of each type of the mover’s pieces (king, queen, rook, bishop, knight, and pawn). A value of 1 indicates the presence of the corresponding piece on a specific square, and 0 indicates absence.
- **6 Channels for Opponent’s Pieces:** Similar encoding for the opponent’s pieces, providing a consistent view from the mover’s perspective.

- **Move Encoding (4 channels):**

- **2 Channels for the Previous Move:** Encoding the starting and ending squares of the last move made, crucial for understanding the transition leading to the current state.
- **2 Channels for the Next Move:** Encoding the anticipated starting and ending squares of the next move, helping the model anticipate the upcoming state and contextualize the puzzle’s solution.

To focus on the solver’s perspective, the board is mirrored if the mover is Black, ensuring consistency in representation.

The encoding process begins with the first position after the initial move that leads to the creation of the puzzle—often a blunder or inaccuracy. From this position, we encode every alternate position in the sequence:

$$\mathbf{P} = \{B_1, B_3, B_5, \dots, B_N\}.$$

By selecting every alternate position, we reduce redundancy while capturing essential changes that reflect the puzzle’s evolution. Each B_n represents a specific state from the solver’s perspective, ensuring the model can learn from decisions made throughout the puzzle-solving process.

B. Model Architecture

Our proposed model, named **GlickFormer**, is designed to predict the difficulty of chess puzzles as measured by their Glicko-2 ratings. GlickFormer effectively captures both spatial and temporal dynamics inherent in the puzzle-solving process. Drawing inspiration from the Vision Transformers for Video (ViViT) architecture [10], which addresses video classification by processing spatial and temporal information in a factorized manner, GlickFormer adapts these principles to handle sequences of chess positions. Each move in a chess puzzle is treated analogously to a frame in a video sequence.

We explore two variants of GlickFormer based on ViViT’s factorization strategies: the *Factorized Encoder GlickFormer* and the *Factorized Self-Attention GlickFormer* (Figure 3). The Factorized Encoder GlickFormer processes spatial features of each chess position using a transformer encoder and then incorporates temporal relationships via a separate temporal transformer encoder. In contrast, the Factorized Self-Attention GlickFormer integrates temporal information directly within the self-attention mechanism by factorizing the attention into